

Practical 3: Time Series Decomposition using STL

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1. Question

Consider the following datasets from the R library and write R code to decompose the time-series data into its trend, seasonal, and residual components. Furthermore, identify the dominating component(s) in the dataset:

- (a) Nottem data
- (b) AirPassengers data

2. Objective

To perform time-series decomposition on the nottem and AirPassengers datasets using the STL (Seasonal and Trend decomposition using Loess) method in R to isolate and identify trend, seasonality, and remainder components.

3. R Code

```
# Clear workspace
rm(list = ls(all = TRUE))

# Load required libraries
library(datasets)
library(TTR)

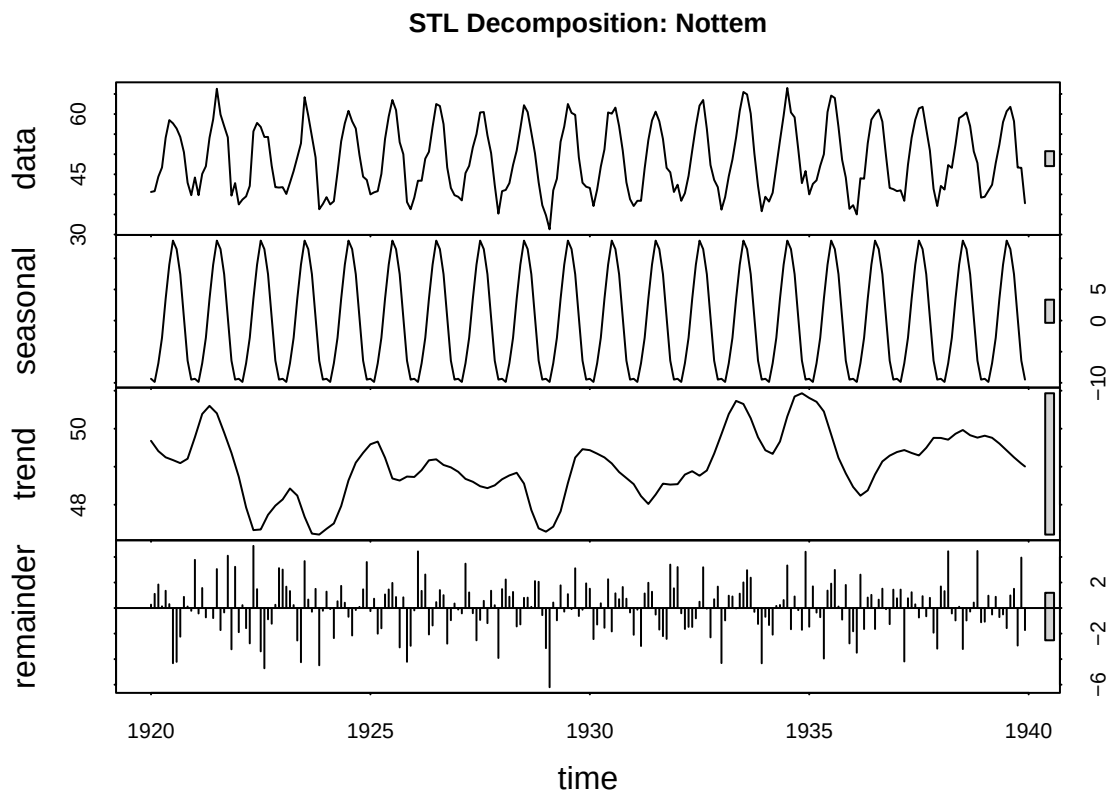
# Load datasets
data(nottem)
data(AirPassengers)

# (a) STL Decomposition for Nottem dataset
nottem_stl <- stl(nottem, s.window = "period")
plot(nottem_stl, main = "STL Decomposition: Nottem")

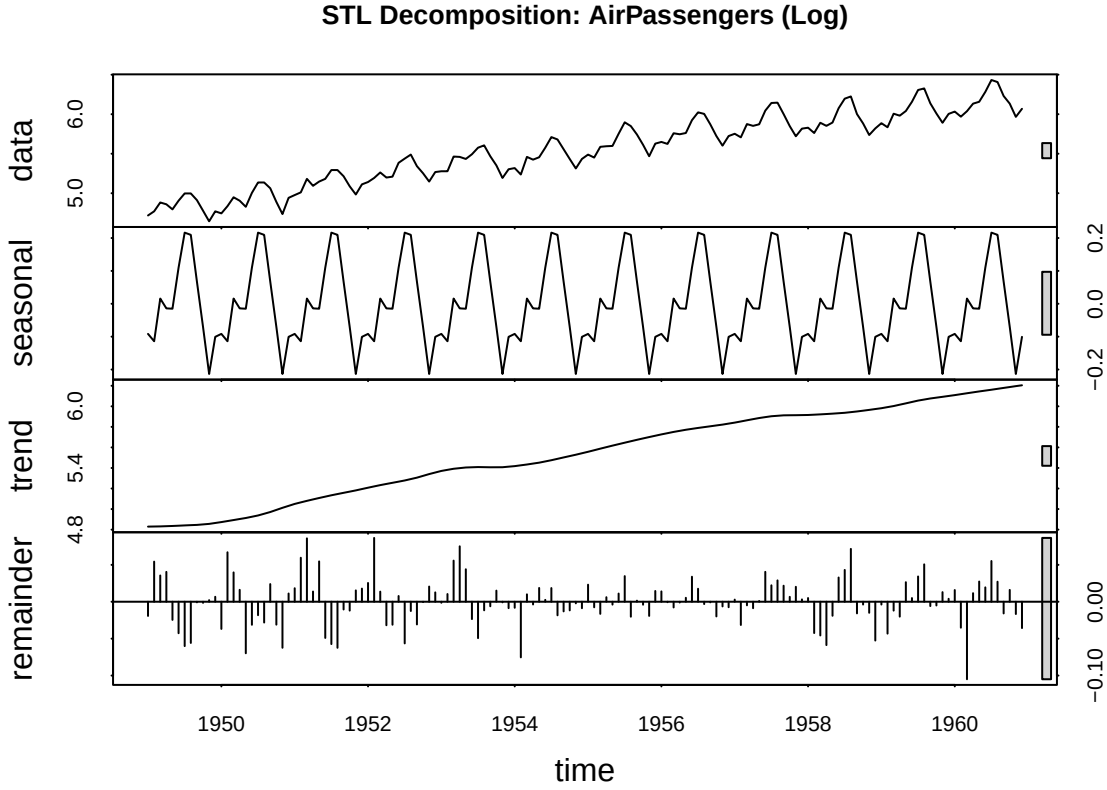
# (b) STL Decomposition for AirPassengers dataset
ap_log <- log(AirPassengers)
ap_stl <- stl(ap_log, s.window = "period")
plot(ap_stl, main = "STL Decomposition: AirPassengers (Log)")
```

4. Output

(a) STL Decomposition: Nottem



(b) STL Decomposition: AirPassengers



5. Conclusion

Through this practical, we successfully decomposed two different types of time-series datasets using the STL method.

For the Notttem dataset, the seasonal component is dominant. The data exhibits strong and consistent yearly seasonal fluctuations, while the trend component remains relatively stable over time.

For the AirPassengers dataset, the trend component is dominant, showing a clear long-term upward movement in passenger numbers. The seasonal component is also strong, but its magnitude increases over time. Applying the logarithmic transformation stabilized the variance before decomposition.

Thus, STL decomposition effectively separates the trend, seasonal, and residual components, allowing clear identification of the dominating patterns in both datasets.